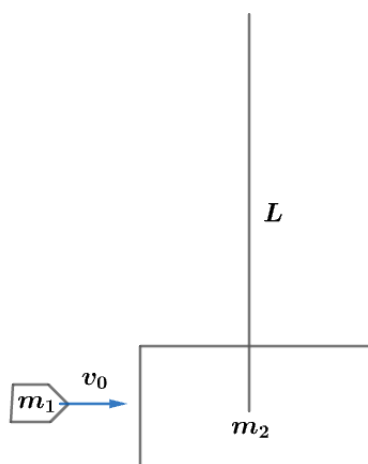


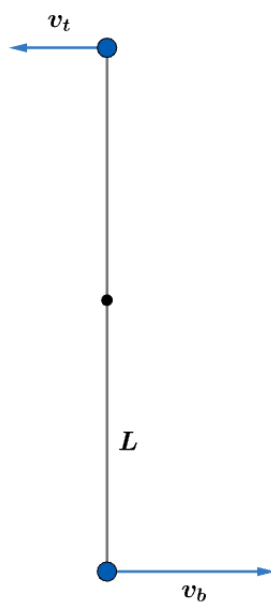
2008 F=ma Exam: Problem 22

Kevin S. Huang



By conservation of linear momentum,

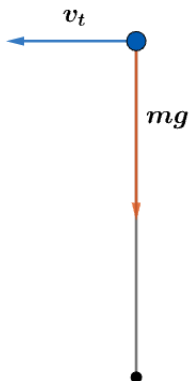
$$m_1 v_0 = (m_1 + m_2) v_b$$
$$v_b = \frac{m_1}{m_1 + m_2} v_0$$



We conserve energy between the bottom and top of the vertical circle,

$$\frac{1}{2}(m_1 + m_2)v_b^2 = \frac{1}{2}(m_1 + m_2)v_t^2 + 2(m_1 + m_2)gL$$

$$v_b^2 = v_t^2 + 4gL$$



The minimum velocity at the top of the vertical circle is attained when the tension in the string goes to zero. Only gravity provides the centripetal acceleration so

$$mg = \frac{mv_t^2}{L}$$

$$v_t^2 = gL$$

Substituting this into the equation for v_b ,

$$v_b^2 = 5gL$$

$$\frac{m_1}{m_1 + m_2}v_0 = \sqrt{5gL}$$

$$v_0 = \frac{(m_1 + m_2)\sqrt{5gL}}{m_1}$$

so the answer is E.